

Testing with one real parameter

Outline

- 1) One-sided tests in general
- 2) Two-sided tests
- 3) UMP unbiased tests

One-sided tests in general

One-param family $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, $\Theta \subseteq \mathbb{R}$

$H_0: \theta \overset{\geq}{\leq} \theta_0$ vs $H_1: \theta \overset{<}{>} \theta_0$ called one-sided hypothesis

$T(X)$ is stochastically increasing in θ if

$P_\theta(T(X) \leq t)$ non-increasing in θ , for all t

Stochastically incr. test stat $\Rightarrow$ calibrate thresh. at θ_0

Prop Suppose $\mathcal{P}$ is one-param family,
 $T(X)$ stochastically increasing

Testing $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$

$$\text{Then } \phi(X) = \begin{cases} 1 & T(X) > c \\ \gamma & T(X) = c \\ 0 & T(X) < c \end{cases}$$

has $\beta(\theta) \nearrow$, is valid iff $\mathbb{E}_{\theta_0} \phi(X) \leq \alpha$

Proof:

$$\begin{aligned} \mathbb{E}_\theta \phi(X) &= P_\theta(T > c) + \gamma P_\theta(T = c) \\ &= (1 - \gamma) P_\theta(T > c) + \gamma P_\theta(T \geq c) \nearrow_{\text{in } \theta} \end{aligned}$$

$$\Rightarrow \sup_{\theta \leq \theta_0} \mathbb{E}_\theta \phi(X) = \mathbb{E}_{\theta_0} \phi(X) \quad \square$$

Score Test

Often, no UMP test exists: LRT depends on θ_1

If n large, could prioritize $\theta_1 = \theta_0 + \varepsilon$, small ε

$$\log LR(x) = \log \frac{p_{\theta_0 + \varepsilon}(x)}{p_{\theta_0}(x)} \approx \varepsilon \cdot \dot{\ell}(\theta_0; x)$$

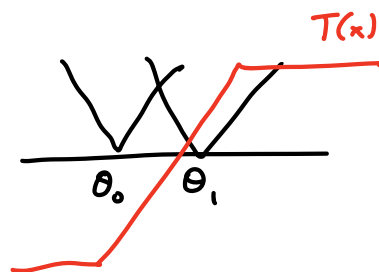
$\Rightarrow$ Use score at θ_0 $S_{\theta_0}(x)$ as test stat.

Ex. Laplace: $X_1, \dots, X_n \stackrel{iid}{\sim} \frac{1}{2} e^{-|x - \theta|}$

Test $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$

$$\begin{aligned} \theta_1 > \theta_0: \log(p_{\theta_1}(x)/p_{\theta_0}(x)) &= \sum_{i=1}^n |x_i - \theta_0| - |x_i - \theta_1| \\ &= \sum T(x_i) \end{aligned}$$

$$T(x) = \begin{cases} \theta_0 - \theta_1 & x \leq \theta_0 \\ 2x - \theta_0 - \theta_1 & \theta_0 \leq x \leq \theta_1 \\ \theta_1 - \theta_0 & x \geq \theta_1 \end{cases}$$



$$\text{Score } S_{\theta_0}(x) = \frac{d}{d\theta} \sum_i -|x_i - \theta| \Big|_{\theta=\theta_0}$$

$$= \sum_{i=1}^n \text{sign}(x_i - \theta_0)$$

called
Sign test

$$= 2 \cdot \# \{x_i > \theta_0\} - n$$

Sign Test

Equivalent: $B(x) = \sum_{i=1}^n 1\{x_i > \theta_0\}$ $= \frac{S_{\theta_0}(x) + n}{2}$ from above

$$\sim \text{Binom}(n, P_{\theta}(x_i > \theta_0))$$

$$\stackrel{\theta = \theta_0}{=} \text{Binom}(n, 1/2)$$

Can interpret as nonparametric test of $H_0: \text{median} \leq \theta_0$

Ex Nonparametric quantile testing

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F \text{ (cdf)}$$

(lower) ξ quantile: $\theta(F) = \min \{t : F(t) \geq \xi\}$ ↙ right-cts

Test $H_0: \theta(F) \leq \theta_0$ vs $H_1: \theta(F) > \theta_0$

$$\Leftrightarrow H_0: F(\theta_0) \geq \xi \text{ vs } H_1: F(\theta_0) < \xi$$

$$B(x) \sim \text{Binom}(n, 1 - F(\theta_0))$$

stoch. smaller $\rightarrow$ $\stackrel{H_0}{\leq} \text{Binom}(n, 1 - \xi)$

Reject for large $B(x)$

Two-sided Alternatives

Setup: $\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$, $\theta_0 \in \Theta^0$

Test $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$

(Can be generalized naturally to $H_0: \theta \in [\theta_1, \theta_2]$)

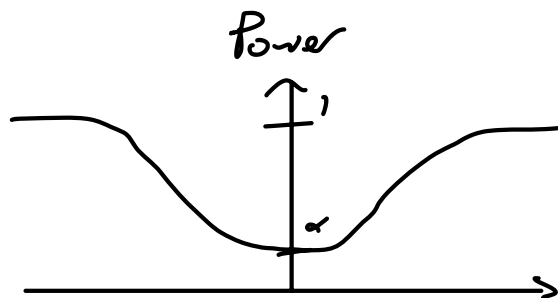
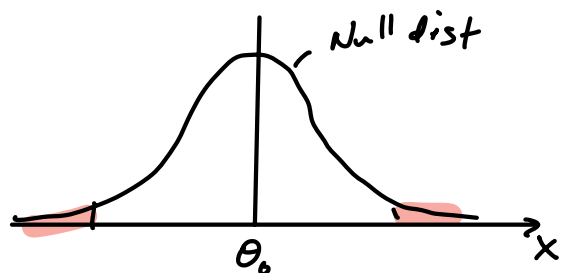
Two-tailed test rejects when $T(X)$ is "extreme"

$$\phi(x) = \begin{cases} 1 & T(X) > c_2 \text{ or } T(X) < c_1 \\ 0 & T(X) \in (c_1, c_2) \\ \gamma_i & T(X) = c_i \end{cases}$$

Two ways to reject. How to balance?

For symmetric distributions like $N(\theta, 1)$,
natural choice is to equalize "lobes" of rej. region

$$\phi_2(x) = 1\{|x - \theta_0| > z_{\alpha/2}\} \text{ for } H_0: \theta = \theta_0$$



For asymmetric dists, or interval null $H_0: \theta \in [\theta_1, \theta_2]$
more complicated

Equal-tailed & unbiased tests

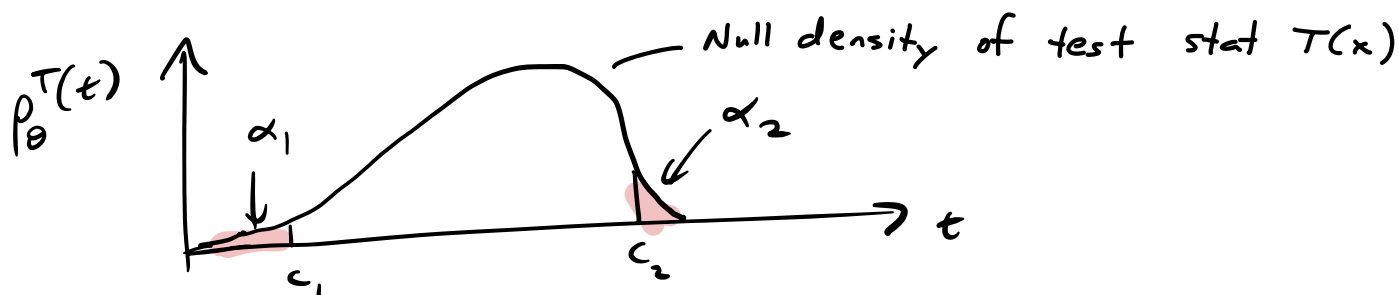
Point null ($H_0: \theta = \theta_0$)

$$\text{Let } \alpha_1 = P_{\theta_0}(T < c_1) + \gamma_1 P_{\theta_0}(T = c_1)$$

$$\alpha_2 = P_{\theta_0}(T > c_2) + \gamma_2 P_{\theta_0}(T = c_2)$$

Valid if $\alpha_1 + \alpha_2 = \alpha$ (α_1 is "free parameter")

Idea 1: Equal-tailed test : $\alpha_1 = \alpha_2 = \frac{\alpha}{2}$



Ex $X \sim \text{Exp}(\theta)$, test $H_0: \theta = 1$

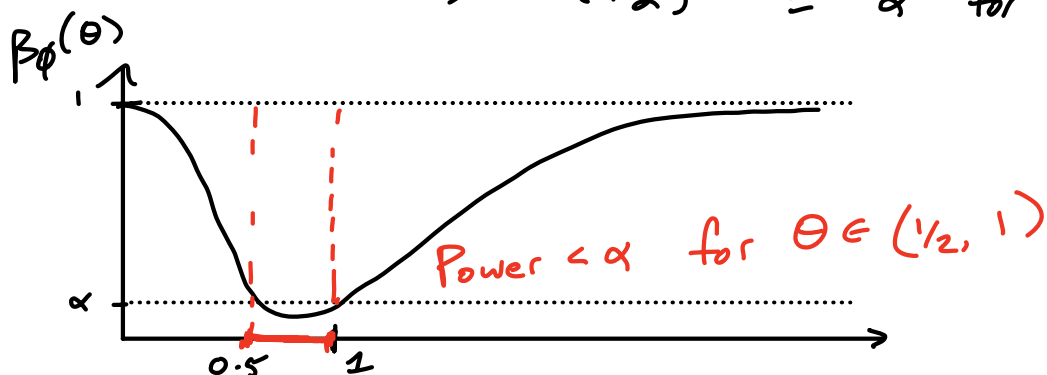
Solve for cutoffs: $\frac{\alpha}{2} = P_1(X \leq c_1) = 1 - e^{-c_1} \Rightarrow c_1 = -\log(1 - \frac{\alpha}{2})$

$$1 - \frac{\alpha}{2} = 1 - e^{-c_2} \Rightarrow c_2 = -\log(\frac{\alpha}{2})$$

$$\phi(x) = 1\{X < -\log(1 - \frac{\alpha}{2})\} + 1\{X > -\log(\frac{\alpha}{2})\}$$

$$\beta_\phi(\theta) = P_\theta\left\{\frac{X}{\theta} < \frac{-\log(1 - \frac{\alpha}{2})}{\theta}\right\} + P_\theta\left\{\frac{X}{\theta} > \frac{-\log(\frac{\alpha}{2})}{\theta}\right\}$$

$$= 1 - (1 - \frac{\alpha}{2})^{1/\theta} + (\frac{\alpha}{2})^{1/\theta} = \alpha \text{ for } \theta = 1 \text{ or } 1/2$$



Unbiased tests

Def $\phi(x)$ is unbiased if $\inf_{\theta \in \Theta} \mathbb{E}_{\theta} \phi(x) \geq \alpha$

Idea 2: Unbiased test: ensure $\min_{\theta} \beta(\theta) = \alpha$

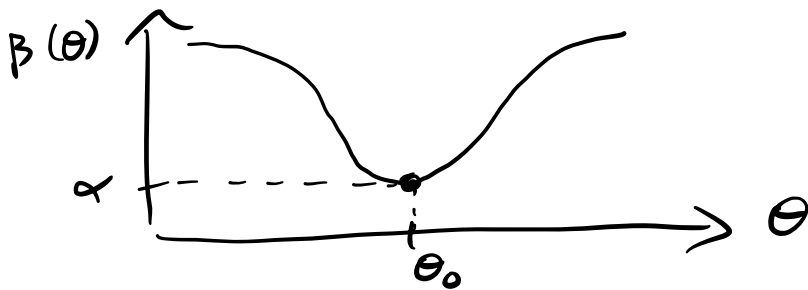
Choose c_1, γ_1 and c_2, γ_2 to solve:

$$\beta(\theta_0) = \alpha \quad (2 \text{ equations, "2" unknowns})$$

$$\dot{\beta}(\theta_0) = 0$$

$$(c_1, \gamma_1) \leftrightarrow \alpha_1$$

$$(c_2, \gamma_2) \leftrightarrow \alpha_2$$



Ex: 1-parameter exp. family, $H_0: \eta = \eta_0$ vs $H_1: \eta \neq \eta_0$

$$X \sim e^{\eta^T(x) - A(\eta)} h(x) \quad (\text{MLR in } T(X))$$

Assume $T(X)$ continuous, solve

$$\alpha = \beta(\eta_0) = \mathbb{P}_{\eta_0}(T < c_1) + \mathbb{P}_{\eta_0}(T > c_2)$$

$$0 = \frac{d}{d\eta} \mathbb{E} \phi(x) = \text{Cov}_{\eta_0}(\phi(T), T)$$

$$= \mathbb{E}_{\eta_0}[(\phi(T) - \alpha) T(x)]$$

Theorem Assume $X \sim e^{\theta T(x) - A(\theta)} h(x)$

$$H_0: |\theta - \theta_0| \leq \delta \quad \text{vs} \quad H_1: |\theta - \theta_0| > \delta, \quad \delta \geq 0$$

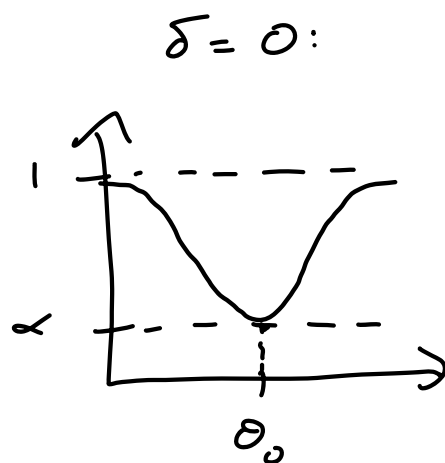
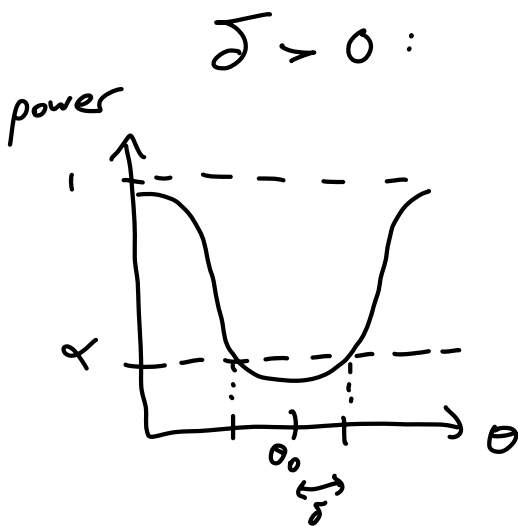
Let ϕ^* be test that rejects for extreme $T(x)$, with $c_1, c_2, \gamma_1, \gamma_2$ chosen so:

$$(i) \beta_{\phi^*}(\theta_0 + \delta) = \beta_{\phi^*}(\theta_0 - \delta) = \alpha$$

and, if $\delta = 0$ (point null)

$$(ii) \quad 0 = \dot{\beta}_{\phi^*}(\theta_0) = \mathbb{E}_{\theta_0}[(T - \mathbb{E}_{\theta_0} T) \phi(x)]$$

Then ϕ^* is UMPU



Proof:

($\delta=0$): Assume wlog $\theta_0 = 0$, $A(0) = 0$

Fix $\theta_1 \neq 0$: want to solve

$$\text{maximize } \int \phi \rho_{\theta_1} d\mu$$

$$\text{s.t. } \int \phi \rho_0 d\mu = \alpha \quad \leftarrow \begin{matrix} \text{(unbiased)} \\ \Rightarrow \text{equality} \end{matrix}$$

$$\int \phi(T - \tau_0) \rho_0 d\mu = 0 \quad (\tau_0 = \mathbb{E}_0 T)$$

Lagrange form:

$$\begin{aligned} \max \quad & \int \phi (\rho_{\theta_1} - \lambda_1 \rho_0 - \lambda_2 \rho_0 (T - \tau_0)) d\mu \\ = \quad & \int \phi \left(\frac{\rho_{\theta_1}}{\rho_0} - \lambda_1 - \lambda_2 (T - \tau_0) \right) dP_0 \end{aligned}$$

$$\text{solutions: } \phi(x) = \begin{cases} 1 & LR(x) > \lambda_1 + \lambda_2 (T - \tau_0) \\ 0 & LR(x) < \lambda_1 + \lambda_2 (T - \tau_0) \\ \text{anything} & LR(x) = \lambda_1 + \lambda_2 (T - \tau_0) \end{cases}$$

Claim: $\phi^*(x)$ is of this form for some λ_1, λ_2

$$LR(x) = e^{\theta_1 T(x) - A(\theta_1)}$$

$$\text{Let } f(t) = e^{\theta_1 t - A(\theta_1)}$$

Set λ_1, λ_2 so $f(c_i) = 0$

$$\lambda_2 = \frac{e^{\theta_1 c_2} - e^{\theta_1 c_1}}{(c_2 - c_1) e^{A(\theta_1)}}$$

$$\lambda_1 = e^{\theta_1 c_1 - A(\theta_1)} - \lambda_2 (c_1 - \tau_0)$$

$f(t)$ convex $\Rightarrow$

$$f(t) < 0 \quad \text{for } c_1 < t < c_2$$

$$f(t) > 0 \quad \text{for } t < c_1 \text{ or } t > c_2$$

$\Rightarrow \phi^*$ maximizes Lagrangian for these $\lambda_1(\theta_1), \lambda_2(\theta_1)$

Suppose ϕ, ϕ^* satisfy constraints,

ϕ^* maximizes Lagrangian for λ_1, λ_2

$$\begin{aligned} \beta_\phi(\theta_1) &= \beta_\phi(\theta_1) + \lambda_1 (\beta_\phi(0) - \alpha) \\ &\quad + \lambda_2 \dot{\beta}_\phi(0) \end{aligned}$$

$$\begin{aligned} &\leq \beta_{\phi^*}(\theta_1) + \lambda_1 (\beta_{\phi^*}(0) - \alpha) \\ &\quad + \lambda_2 \dot{\beta}_{\phi^*}(0) \end{aligned}$$

$$= \beta_{\phi^*}(\theta_1)$$

But θ_1 was arbitrary

($\delta > 0$) Assume $\theta_0 + \delta = 0$, $\theta_0 - \delta = -1$, $A(0) = 0$

Fix $\theta_1 > 0$, want to solve

$$\max \int \phi \rho_{\theta_1} d\mu$$

$$\text{s.t. } \int \phi \rho_0 d\mu = \int \phi \rho_{-1} d\mu = \alpha$$

Lagrangian:

$$\begin{aligned} & \int \phi (\rho_{\theta_1} - \lambda_1 \rho_0 - \lambda_2 \rho_{-1}) d\mu \\ &= \int \phi \left(\frac{\rho_{\theta_1}}{\rho_0} - \lambda_1 - \lambda_2 \frac{\rho_{-1}}{\rho_0} \right) dP_0 \\ &= \int \phi \left(e^{\theta_1 T - A(\theta_1)} - \lambda_1 - \lambda_2 e^{-T - A(-1)} \right) dP_0 \\ &= e^{-A(\theta_1)} \int \phi \left(e^{\theta_1 T} - \gamma_1 - \gamma_2 e^{-T} \right) dP_0 \end{aligned}$$

$$\text{for } \gamma_1 = \lambda_1 e^{A(\theta_1)} \quad \gamma_2 = \lambda_2 e^{A(\theta_1) - A(-1)}$$

$$\text{Set } \gamma_2 = \frac{e^{\theta_1 c_2} - e^{\theta_2 c_1}}{e^{-c_2} - e^{-c_1}} < 0$$

$$\gamma_1 = e^{\theta_1 c_1} - \gamma_2 e^{-c_1}$$

$$f(t) = e^{\theta_1 t} - \gamma_1 - \gamma_2 e^{-t} \quad \text{convex since } \gamma_2 < 0$$

$$f(c_1) = f(c_2) = 0 \quad \text{by construction}$$

Rest of proof same $\square$